HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed backend services in **Java 11/17 and Spring Boot** (REST APIs, microservices) for a banking client's payments-operations dashboard, as stateless services that **scale horizontally** on AWS.
- Wrote **multithreaded Java** code (ExecutorService, CompletableFuture, concurrent collections) to batch-process payment messages, applying **parallel computing** principles with thread-safe shared state.
- Investigated **system performance and memory issues** in production Java services using heap and thread dumps, GC logs and **Dynatrace**, and fixed memory leaks and slow endpoints.
- Applied **object-oriented design and design patterns** (Strategy, Factory, Builder) when refactoring legacy Java modules, making them easier to test, extend and support.
- Built data access layers with **Hibernate/JPA and SQL** on PostgreSQL, Oracle and MySQL; tuned slow queries and indexes and supported **data migration** between systems during integrations.
- Secured REST APIs with **Spring Security, OAuth2 and JWT**, implementing role-based **authentication and authorization** for internal users and service-to-service calls.
- Supported production as part of the **on-call rotation** for P1/P2 incidents: triaged alerts in Splunk and CloudWatch, led **root-cause analysis** and delivered permanent fixes to stop repeat issues.
- Worked across the **full software development life cycle**: coding standards, **code reviews**, Git, Maven builds, Jenkins CI/CD, **JUnit and Mockito** tests, and Docker deployments to AWS (EC2, S3, Lambda).

SKILLS

Java – Java 11/17, OOP, multithreading & concurrency, JVM memory management, design patterns.

Frameworks – Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Hibernate, JUnit, Mockito.

Backend – REST APIs, microservices, distributed systems, OAuth2/JWT, caching (Redis), horizontal scaling.

Databases – PostgreSQL, Oracle, MySQL; SQL query tuning, indexing, data migration.

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins CI/CD, Maven, Git, Linux.

Operations – On-call production support (P1/P2), root-cause analysis, Splunk, Dynatrace, CloudWatch.

Other Languages – Python, SQL, Bash.

PROJECTS

Distributed Caching Layer for a Spring Boot Service (Personal Project)

- Built a Java/Spring Boot REST service with a **Redis** cache-aside layer (TTL-based eviction) to avoid repeated database reads, and compared response times with and without the cache using JMeter.
- Ran Redis as a primary with a replica and **Sentinel failover** in Docker Compose to learn **replication** and high availability, testing what happens when the primary node goes down.
- Spread keys across several Redis nodes with **consistent hashing** to explore **sharding** and adding nodes (horizontal scaling) with minimal key movement.
- Secured the endpoints with Spring Security and JWT, and covered the service with JUnit, Mockito and Testcontainers integration tests.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)